



USL Assignment 5

Multidimensional Scaling Dimensionality Reduction Technique

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MDS THEORY

1. Introduction to Multidimensional Scaling (MDS)

Multidimensional Scaling (MDS) is a dimensionality reduction technique used in machine learning and data analysis to represent high-dimensional data in a lower-dimensional space while preserving the pairwise distances between data points.

Similar to Singular Value Decomposition (SVD), MDS is an unsupervised learning technique, meaning it does not require target labels and focuses only on the structure of the data.

The main goal of MDS is to create a spatial representation of data such that the distances between points in the reduced space closely match the original distances in the high-dimensional space.



This makes MDS especially useful for:

- Data visualization
- Pattern recognition
- Understanding similarity between data points

2. Mathematical Concept Behind MDS

MDS works by analyzing the **distance matrix** of the dataset.

For a dataset with n points, a distance matrix D is constructed:

$$D = \begin{bmatrix} 0 & d_{12} & d_{13} \\ d_{21} & 0 & d_{23} \\ d_{31} & d_{32} & 0 \end{bmatrix}$$

Where:

- d_{ij} = distance between point i and j

Objective of MDS

MDS finds a new set of points in lower dimensions such that:

$$d_{ij} \approx d'^{ij}$$



Where:

- d_{ij} = original distance
- d^{ij} = distance in reduced space

3. Stress Function (Quality Measure)

To evaluate how well distances are preserved, MDS uses the **Normalized Stress Function**:

$$Stress_1^2 = \frac{\sum_{i,j} (d_{ij} - \tilde{d}_{ij})^2}{\sum_{i,j} d_{ij}^2}$$

4. Steps of the MDS Algorithm

The main steps involved in applying MDS are:

1. Compute the pairwise distance matrix of the dataset
2. Initialize points randomly in lower-dimensional space
3. Compute distances in reduced space
4. Compare with original distances
5. Minimize the stress function iteratively
6. Obtain optimized low-dimensional representation



5. Example of MDS

Consider three points in high-dimensional space:

- A, B, C

Their distances are:

- $D_{ab}=5$
- $D_{ac}=7$
- $D_{bc}=3$

MDS maps these points into 2D such that these distances are approximately preserved.

Thus, even after dimensionality reduction, the relative positioning of points remains meaningful.

6. MDS Applied to Dataset (Penguins Dataset)

The Penguins dataset contains physical measurements of penguins such as:

- Bill length
- Bill depth
- Flipper length



- Body mass

This dataset forms a **multi-dimensional feature space**, similar to how image datasets have many pixel features.

MDS is applied to:

- Reduce dimensionality from 4D $\rightarrow$ 2D
- Preserve similarity between penguins
- Visualize species clusters

After applying MDS:

- Data is transformed into 2D space
- Clusters of species can be visually analyzed

7. Inferences from MDS

After applying Multidimensional Scaling:

- The dataset is reduced to lower dimensions while preserving pairwise distances
- The stress value indicates the quality of embedding
- A low stress value means distances are well preserved
- However, classification accuracy may decrease due to loss of feature-specific information

This shows that:

- MDS is excellent for **visualization**



- But not always optimal for **classification performance**

8. References

- <https://scikit-learn.org/stable/modules/generated/sklearn.manifold.MDS.html>
- <https://www.geeksforgeeks.org/multidimensional-scaling/>
- [Borg & Groenen – Modern Multidimensional Scaling](#)